

理论物理基础 1-电动力学

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1 相对论性分析力学

协变的表述:

定义逆变矢量

$$x^\mu = (ct, x, y, z)$$

定义协变矢量

$$x_\mu = (ct, -x, -y, -z)$$

通过度规联系:

$$g_{\mu\nu} = \text{diag}\{1, -1, -1, -1\}$$

$$\Rightarrow x_\mu = g_{\mu\nu} x^\nu$$

自由粒子运动

定义作用量:

$$S = -mc \int \sqrt{dx_\mu dx^\mu}$$

最小作用量原理:

$$\begin{aligned} \delta S &= -mc \int \frac{\delta(dx_\mu dx^\mu)}{2\sqrt{dx_\mu dx^\mu}} = -mc \int \frac{dx_\mu \delta(dx^\mu)}{ds} \\ &= mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds = 0 \\ &\Rightarrow \frac{d^2 x_\mu}{ds^2} = 0 \end{aligned}$$

考虑原时:

$$ds = cd\tau \Rightarrow \frac{d^2 x_\mu}{d\tau^2} = 0$$

可以定义四速度:

$$\frac{dx_\mu}{d\tau} = \gamma(c, -\vec{v})$$

四动量:

$$p_\mu = m \frac{dx_\mu}{d\tau} = \left(\frac{E}{c}, -\vec{p} \right)$$

带电粒子在电磁场中的运动

定义作用量:

$$S = -mc \int ds - e \int A_\mu dx^\mu$$

四维势

$$A_\mu = \left(\frac{\phi}{c}, -\vec{A} \right)$$

电磁场

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

变分:

$$A_\mu d(\delta x^\mu) = -\frac{\partial A_\mu}{\partial x^\nu} dx^\nu \delta x^\mu$$

$$\delta A_\mu dx^\mu = \frac{\partial A_\mu}{\partial x^\nu} \delta x^\nu dx^\mu$$

$$\Rightarrow \delta A_\mu dx^\mu = \int \left(\frac{\partial A_\nu}{\partial x^\mu} \frac{dx^\nu}{ds} - \frac{\partial A_\mu}{\partial x^\nu} \frac{dx^\nu}{ds} \right) \delta x^\mu ds = 0$$

$$\Rightarrow mc \frac{d^2 x_\mu}{ds^2} + e \left(\frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu} \right) \frac{dx^\nu}{ds} = 0$$

定义电磁场张量:

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^\nu}$$

$$\Rightarrow m \frac{d^2 x_\mu}{d\tau^2} = e F_{\mu\nu} u^\nu$$

电磁场张量:

$$F_{\mu\nu} = \begin{pmatrix} 0 & \frac{E_x}{c} & \frac{E_y}{c} & \frac{E_z}{c} \\ -\frac{E_x}{c} & 0 & -B_z & B_y \\ -\frac{E_y}{c} & B_z & 0 & -B_x \\ -\frac{E_z}{c} & -B_y & B_x & 0 \end{pmatrix}$$

可以给出运动方程:

$$\frac{d}{dt} (\gamma mc^2) = e \vec{E} \cdot \vec{v}$$

$$\frac{d\vec{p}}{dt} = e \vec{E} + e (\vec{v} \times \vec{B})$$

场的运动方程

拉格朗日量密度

$$\mathcal{L} = \frac{F_{\mu\nu} F^{\mu\nu}}{4\mu_0} + J^\mu A_\mu$$

$$J^\mu = \frac{\rho}{\gamma} u^\mu = (\rho c, \rho \vec{v}) = (\rho c, \vec{j})$$

变分:

$$\delta S = \int \left(\mu_0 J^\mu + \frac{\partial F^{\mu\nu}}{\partial x^\nu} \right) \mu_0 \delta A_\mu c dt d^3x$$

$$\Rightarrow \frac{\partial F^{\mu\nu}}{\partial x^\nu} = -\mu_0 J^\mu$$

取 $\mu = 0$

$$\Rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

取 $\mu = 1, 2, 3$

$$\Rightarrow \nabla \times \vec{B} = \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{j}$$

比安基恒等式:

$$\frac{\partial F_{\nu\rho}}{\partial x^\mu} + \frac{\partial F_{\rho\mu}}{\partial x^\nu} + \frac{\partial F_{\mu\nu}}{\partial x^\rho} = 0$$

取 $(\mu, \nu, \rho) = (1, 2, 3)$

$$\Rightarrow \nabla \cdot \vec{B} = 0$$

取 $(\mu, \nu, \rho) = (0, 1, 2) (0, 1, 3) (0, 2, 3)$

$$\Rightarrow \nabla \times \vec{B} = -\frac{\partial \vec{E}}{\partial t}$$

推导出了完整的 Maxwell 方程组。

这里相关的推导过程完全取决于电磁场张量是如何定义的，如果电磁场张量定义差了负号，相应的推导也要变化。

2 经典电动力学基础

2.1 基本公式

电场强度:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \iiint \frac{\rho(\vec{r}') (\vec{r} - \vec{r}') d^3r'}{|\vec{r} - \vec{r}'|^3}$$

电势:

$$\phi = \frac{1}{4\pi\varepsilon_0} \iiint \frac{\rho(\vec{r}') d^3r'}{|\vec{r} - \vec{r}'|}$$

磁感应强度:

$$\vec{B} = \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}(\vec{r}') \times (\vec{r} - \vec{r}') d^3r'}{|\vec{r} - \vec{r}'|^3}$$

磁矢势:

$$\vec{A} = \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}(\vec{r}') d^3r'}{|\vec{r} - \vec{r}'|}$$

其中 $\vec{r}'$ 为源点, $\vec{r}$ 为场点, 对源积分。

2.2 介质中的 Maxwell 方程组

束缚电荷密度:

$$\rho_b = -\nabla \cdot \vec{P}$$

束缚电荷面密度:

$$\Sigma_b = \vec{P} \cdot \hat{n}$$

磁化电流密度:

$$\vec{j}_m = \nabla \times \vec{M}$$

磁化电流面密度:

$$\vec{K}_m = \vec{M} \times \hat{n}$$

边值关系:

$$\hat{n} \cdot (\vec{D}_2 - \vec{D}_1) = \Sigma_f$$

$$\hat{n} \cdot (\vec{B}_2 - \vec{B}_1) = 0$$

$$\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}_f$$

$$\hat{n} \times (\vec{E}_2 - \vec{E}_1) = 0$$

2.3 电磁场中的守恒量

1、电荷守恒

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

proof:

$$Q = \int \rho d^3r$$

$$\frac{dQ}{dt} = - \oint \vec{j} \cdot \vec{\sigma}$$

数学公式:

$$\int \frac{\partial \rho}{\partial t} d^3r = - \int \nabla \cdot \vec{j} d^3r$$

$$\Rightarrow \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

2、能量守恒

能量密度:

$$u = \frac{1}{2} \left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0} \right)$$

能流密度:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

能量关系:

$$\iiint d^3r \left(\frac{\partial u}{\partial t} + \vec{j} \cdot \vec{E} + \nabla \cdot \vec{S} \right) = 0$$

体积内电磁场能量的增加 = 通过表面流入的能流-电场对电荷做功所消耗的能量。

proof:

已知能量关系

$$\frac{dW}{dt} = \int \vec{j} \cdot \vec{E} d^3r$$

由 Maxwell 方程:

$$\begin{aligned} \nabla \times \vec{B} &= \mu_0 \vec{j} + \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} \\ \Rightarrow \vec{j} &= \frac{1}{\mu_0} \nabla \times \vec{B} - \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \\ \Rightarrow \frac{dW}{dt} &= \int \left[\frac{1}{\mu_0} (\vec{E} \cdot \nabla \times \vec{B}) - \varepsilon_0 \vec{E} \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) \right] d^3r \end{aligned}$$

由数学公式:

$$\begin{aligned} \nabla \cdot (\vec{E} \times \vec{B}) &= \vec{B} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{B}) \\ \Rightarrow \frac{dW}{dt} &= \int \left\{ \frac{1}{\mu_0} [\vec{B} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{B})] - \varepsilon_0 \vec{E} \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) \right\} d^3r \end{aligned}$$

再由 Maxwell 方程:

$$\begin{aligned} \vec{B} \cdot (\nabla \times \vec{E}) &= -\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \\ \Rightarrow \frac{dW}{dt} &= \int \left[-\frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) - \frac{1}{\mu_0} \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} - \varepsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right] d^3r \\ &= -\frac{\partial}{\partial t} \int \left(\frac{\vec{B}^2}{2\mu_0} + \frac{\varepsilon_0 \vec{E}^2}{2} \right) d^3r - \int \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) d^3r \\ &= -\frac{\partial}{\partial t} \int \left(\frac{\vec{B}^2}{2\mu_0} + \frac{\varepsilon_0 \vec{E}^2}{2} \right) d^3r - \oint \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \cdot d\vec{\sigma} \end{aligned}$$

定义能量密度:

$$u = \frac{\vec{B}^2}{2\mu_0} + \frac{\varepsilon_0 \vec{E}^2}{2}$$

能流密度:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$\Rightarrow \frac{dW}{dt} = -\frac{\partial}{\partial t} \int u d^3r - \oint \vec{S} \cdot \vec{\sigma}$$

或者可以可作

$$\int d^3r \left(\frac{\partial u}{\partial t} + \vec{j} \cdot \vec{E} + \nabla \cdot \vec{S} \right) = 0$$

3、动量守恒

动量密度:

$$\vec{g} = \varepsilon_0 \vec{E} \times \vec{B} = \frac{1}{c^2} \vec{S}$$

协强张量:

$$T_{ij} = \varepsilon_0 \left[E_i E_j + c^2 B_i B_j - \frac{1}{2} (E^2 + c^2 B^2) \delta_{ij} \right]$$

动量关系:

$$\vec{f} + \frac{\partial \vec{g}}{\partial t} = \nabla \cdot \vec{T}$$

机械动量的增加 = 通过应力流入的动量 + 场动量减少所释放的动量。

proof:

洛伦兹力公式:

$$\vec{f} = \rho \vec{E} + \vec{j} \times \vec{B}$$

由 Maxwell 方程:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} + \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \vec{f} = \varepsilon_0 \left(\nabla \cdot \vec{E} \right) \vec{E} + \left(\frac{1}{\mu_0} \nabla \times \vec{B} - \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \times \vec{B}$$

由 Maxwell 方程组和数学公式:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\frac{\partial}{\partial t} \left(\vec{E} \times \vec{B} \right) = \frac{\partial \vec{E}}{\partial t} \times \vec{B} + \vec{E} \times \frac{\partial \vec{B}}{\partial t}$$

$$\Rightarrow \frac{\partial \vec{E}}{\partial t} \times \vec{B} = \frac{\partial}{\partial t} \left(\vec{E} \times \vec{B} \right) + \vec{E} \times \left(\nabla \times \vec{E} \right)$$

$$\Rightarrow \vec{f} = \varepsilon_0 \left(\nabla \cdot \vec{E} \right) \vec{E} + \frac{1}{\mu_0} \left(\nabla \times \vec{B} \right) \times \vec{B} - \varepsilon_0 \frac{\partial}{\partial t} \left(\vec{E} \times \vec{B} \right) - \varepsilon_0 \vec{E} \times \left(\nabla \times \vec{E} \right)$$

由数学公式:

$$\nabla \left(\vec{E}^2 \right) = 2 \left(\vec{E} \cdot \nabla \right) \vec{E} + 2 \vec{E} \times \left(\nabla \times \vec{E} \right)$$

$$\nabla \left(\vec{B}^2 \right) = 2 \left(\vec{B} \cdot \nabla \right) \vec{B} + 2 \vec{B} \times \left(\nabla \times \vec{B} \right)$$

$$\begin{aligned}\vec{f} &= \varepsilon_0 \left(\nabla \times \vec{E} \right) \vec{E} - \frac{1}{\mu_0} \left[\frac{1}{2} \nabla \left(\vec{B}^2 \right) - \left(\vec{B} \cdot \nabla \right) \vec{B} \right] - \varepsilon_0 \left[\frac{1}{2} \nabla \left(\vec{E}^2 \right) - \left(\vec{E} \cdot \nabla \right) \vec{E} \right] - \varepsilon_0 \frac{\partial}{\partial t} \left(\vec{E} \times \vec{B} \right) \\ &= \varepsilon_0 \left[\left(\nabla \cdot \vec{E} \right) \vec{E} + \left(\vec{E} \cdot \nabla \right) \vec{E} \right] + \frac{1}{\mu_0} \left[\left(\nabla \cdot \vec{B} \right) \vec{B} + \left(\vec{B} \cdot \nabla \right) \vec{B} \right] - \nabla \left(\frac{1}{2} \varepsilon_0 E^2 + \frac{B^2}{2\mu_0} \right) - \varepsilon_0 \frac{\partial}{\partial t} \left(\vec{E} \times \vec{B} \right)\end{aligned}$$

引入 Maxwell 应力张量：

$$T_{ij} = \varepsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right)$$

定义动量密度：

$$\begin{aligned}g &= \frac{1}{c^2} \vec{S} \\ \Rightarrow \vec{f} + \frac{\partial \vec{g}}{\partial t} &= \nabla \cdot \vec{T}\end{aligned}$$

3 静电问题

基本方程：

$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\varepsilon} \\ \nabla \times \vec{E} &= 0 \\ \vec{E} &= -\nabla \phi \Rightarrow \nabla^2 \phi = -\frac{\rho}{\varepsilon}\end{aligned}$$

界面关系：

$$\begin{aligned}\phi_1 &= \phi_2 \\ \vec{n} \cdot (\varepsilon_1 \nabla \phi_1 - \varepsilon_2 \nabla \phi_2) &= \Sigma_f\end{aligned}$$

3.1 静电势的多级展开

$$\phi = \frac{1}{4\pi\varepsilon_0} \iiint \frac{\rho(\vec{r}') d^3 r'}{|\vec{r} - \vec{r}'|}$$

展开：

$$\frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{r} + \frac{\vec{r}' \cdot \vec{e}_r}{r^2} + \frac{\vec{r}' \vec{r}' : (3\vec{e}_r \vec{e}_r - \vec{I})}{2r^3}$$

1、第一项：

$$\begin{aligned}\phi_0 &= \frac{1}{4\pi\varepsilon} \iiint \frac{\rho(\vec{r}') d^3 r'}{r} = \frac{Q}{4\pi\varepsilon_0 r} \\ Q &= \iiint \rho(\vec{r}') d^3 r'\end{aligned}$$

2、第二项：

$$\phi_1 = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(\vec{r}') \vec{r}' d^3r' \cdot \vec{e}_r}{r^2} = \frac{1}{4\pi\epsilon} \frac{\vec{p} \cdot \vec{e}_r}{r^2}$$

$$\vec{p} = \iiint \rho(\vec{r}') \vec{r}' d^3r'$$

离散化：

$$\vec{p} = \sum_n q_n \vec{r}_n$$

3、第三项：

$$\phi_2 = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(\vec{r}') \vec{r}' \vec{r}' d^3r' : (3\vec{e}_r \vec{e}_r - \vec{I})}{2r^3}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\vec{D} : (3\vec{e}_r \vec{e}_r - \vec{I})}{2r^3} = \frac{1}{4\pi\epsilon_0} \frac{\vec{D} : \vec{e}_r \vec{e}_r}{2r^3}$$

$$\vec{D} = \iiint \rho(\vec{r}') \vec{r}' \vec{r}' d^3r'$$

$$\vec{D} = 3\vec{D} - Tr(\vec{D}) \vec{I}$$

离散化：

$$\vec{D} = \sum_n q_n \vec{r}_n \vec{r}_n$$

$$\vec{D} = 3\vec{D} - Tr(\vec{D}) \vec{I}$$

3.2 多极子能量

考虑的能量项为：

$$\iiint \epsilon \phi_{\text{外}} \nabla \cdot \vec{E} d^3r' = \iiint \rho(\vec{r}') \phi_{\text{外}}(\vec{r}') d^3r'$$

将电势展开（不写“外”字了）

$$\phi(\vec{r}') = \phi(0) + \vec{r}' \cdot \nabla \phi(0) + \frac{1}{2} \vec{r}' \vec{r}' : \nabla \nabla \phi(0)$$

1、第一项

$$u_0 = \iiint \rho(\vec{r}') \phi(0) d^3r' = Q\phi(0)$$

2、第二项

$$u_1 = \iiint \rho(\vec{r}') \vec{r}' \cdot \nabla \phi(0) d^3r' = \vec{p} \cdot \nabla \phi(0) = -\vec{p} \cdot \vec{E}$$

电偶极子受力

$$\vec{F} = -\nabla u_1 = \nabla (\vec{p} \cdot \vec{E}) = \vec{p} \cdot \nabla \vec{E}$$

受力矩

$$\vec{N} = \vec{p} \times \vec{E}$$

3、第三项

$$u_2 = \iiint \frac{1}{2} \rho(\vec{r}') \vec{r}' \vec{r}' d^3 r' : \nabla \nabla \phi(0) = \frac{1}{2} \vec{D} : \nabla \nabla \phi(0) = \frac{1}{6} \vec{D} : \nabla \nabla \phi(0)$$

3.3 静电边值问题

唯一性定理：已知 $\rho(\vec{r}), \varepsilon$ ，若 $\phi|_s$ 或者 $\frac{\partial \phi}{\partial n}|_s$ 已知，则 $\phi(V_{\text{内}})$ 唯一。

证明需要的数学公式：

$$\nabla(\Phi \nabla \Phi) = \Phi \nabla^2 \Phi + (\nabla \Phi)^2$$

解题方法：1、静电镜像法；2、特解叠加法

4 静磁问题

基本方程：

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

$$\vec{B} = \nabla \times \vec{A}$$

稳定电流：

$$\nabla \cdot \vec{j} = 0$$

4.1 电流环

考虑其电流密度：

$$\vec{j}(\vec{r}') = \frac{I}{a} \delta(r' - a) \delta\left(\theta - \frac{\pi}{2}\right) \vec{e}_\phi$$

磁矢势：

$$\begin{aligned} \vec{A} &= \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3 r' = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{\vec{e}'_\phi d\phi'}{|\vec{r} - a\vec{e}'_r|} \\ A_\phi &= \frac{\mu_0 I a}{4\pi} \int \frac{\cos \phi' d\phi'}{(r^2 + a^2 - 2ra \sin \theta \cos \phi')^{1/2}} \end{aligned}$$

$r \gg a$ 时，展开

$$\frac{1}{(r^2 + a^2 - 2ra \sin \theta \cos \phi')^{1/2}} = \frac{1}{r} + \frac{a \sin \theta \cos \phi'}{r^2}$$

$$\Rightarrow A_\phi = \frac{\mu_0 I S \sin \theta}{4\pi r^2}$$

定义磁矩：

$$\begin{aligned}\vec{m} &= IS\hat{n} \\ \Rightarrow \vec{A} &= \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{e}_r}{r^2}\end{aligned}$$

磁感应强度

$$\vec{B} = \nabla \times \vec{A} = \frac{\mu_0}{4\pi} \left[\vec{m} \nabla \cdot \frac{\vec{e}_r}{r^2} - (\vec{m} \cdot \nabla) \frac{\vec{e}_r}{r^2} \right] = \frac{\mu_0}{4\pi} \frac{\vec{m} \cdot (3\vec{e}_r\vec{e}_r - \vec{I})}{r^3}$$

或者可定义磁标势：

$$\begin{aligned}\vec{B} &= -\nabla \phi_m \\ \phi_m &= \frac{\mu_0}{4\pi} \frac{\vec{m} \cdot \vec{e}_r}{r^2}\end{aligned}$$

4.2 磁多极子展开

$$\vec{A} = \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}(\vec{r}') d^3r'}{|\vec{r} - \vec{r}'|}$$

展开：

$$\frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{r} + \frac{\vec{r}' \cdot \vec{e}_r}{r^2}$$

1、第一项

$$\vec{A}_0 = \frac{\mu_0}{4\pi r} \iiint \vec{j}(\vec{r}') d^3r' = 0$$

2、第二项

$$\vec{A}_1 = \frac{\mu_0}{4\pi r^2} \cdot \iiint \vec{r}' \vec{j}(\vec{r}') d^3r'$$

利用数学公式：

$$\nabla' \cdot (\vec{j}(\vec{r}') \vec{r}' \vec{r}') = (\nabla' \cdot \vec{j}) \vec{r}' \vec{r}' + \vec{j} \vec{r}' + \vec{r}' \vec{j}$$

右边第一项为零，对全空间积分左边也是零：

$$\Rightarrow \iiint (\vec{j} \vec{r}' + \vec{r}' \vec{j}) d^3r' = 0$$

所以

$$\begin{aligned}\vec{A}_1 &= \frac{\mu_0}{4\pi} \frac{\vec{e}_r}{r^2} \iiint (\vec{r}' \vec{j} - \vec{j} \vec{r}') d^3r' \\ &= -\frac{\mu_0}{4\pi} \frac{1}{r^2} \vec{e}_r \times \iiint (\vec{r}' \times \vec{j}) d^3r'\end{aligned}$$

定义磁矩：

$$\vec{m} = \frac{1}{2} \iiint (\vec{r}' \times \vec{j}) d^3 r'$$

$$\Rightarrow \vec{A}_1 = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{e}_r}{r^2}$$

4.3 磁偶极子受力和力矩

$$\vec{F} = \iiint \vec{j} \times \vec{B} d^3 r'$$

磁场展开：

$$\vec{B} = \vec{B}_0 + \vec{r}' \cdot \nabla \vec{B}$$

1、第一项

$$\iiint \vec{j} \times \vec{B}_0 d^3 r' = 0$$

2、第二项

$$\iiint \vec{j} \times (\vec{r}' \cdot \nabla \vec{B}_0) d^3 r' = \iiint (\vec{j} \vec{r}' \cdot \nabla) \times \vec{B}_0 d^3 r'$$

利用前面已经推导过的公式：

$$\iiint \vec{j} \vec{r}' d^3 r' = \frac{1}{2} \iiint (\vec{j} \vec{r}' - \vec{r}' \vec{j}) d^3 r'$$

$$(\vec{j} \vec{r}' - \vec{r}' \vec{j}) \cdot \nabla = (\vec{r}' \times \vec{j}) \times \nabla$$

$$\Rightarrow \vec{F} = (\vec{m} \times \nabla) \times \vec{B}_0 = \nabla (\vec{m} \cdot \vec{B}_0)$$

势能：

$$U = -\vec{m} \cdot \vec{B}_0$$

力矩：

$$\vec{N} = \iiint \vec{r}' \times (\vec{j} \times \vec{B}) d^3 r' = \vec{m} \times \vec{B}_0$$

5 电磁波的传播

5.1 平面电磁波的特性

满足条件：

$$\rho = 0, \vec{j} = 0$$

得到波动方程:

$$\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

$$\nabla^2 \vec{B} - \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

$$\Rightarrow \vec{E} = \vec{E}_0 e^{i\vec{k} \cdot \vec{x} - i\omega t}$$

色散关系:

$$\frac{\omega}{c} = k$$

横波:

$$\vec{k} \cdot \vec{E} = 0$$

$$\vec{k} \cdot \vec{B} = 0$$

$$\vec{E} \cdot \vec{B} = 0$$

电场和磁场关系:

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$$\omega \vec{E} = -c^2 \vec{k} \times \vec{B}$$

偏振特性:

$$Re(E_x) = |E_x| \cos(kz - \omega t + \phi_x)$$

$$Re(E_y) = |E_y| \cos(kz - \omega t + \phi_y)$$

1、 ϕ_x, ϕ_y 一般取值 $\Rightarrow$ 椭圆偏振;

2、 $\phi_x = \phi_y + n\pi \Rightarrow$ 线偏振;

3、 $\phi_x = \phi_y \pm \frac{\pi}{2} \Rightarrow$ 圆偏振

$$\phi_y - \phi_x = \frac{\pi}{2} \Rightarrow \text{左旋}$$

$$\phi_y - \phi_x = -\frac{\pi}{2} \Rightarrow \text{右旋}$$

两组基本的圆偏振模式也可以构成一组基:

$$\vec{e}_{\text{左}} = \frac{1}{\sqrt{2}} (\vec{e}_x + i\vec{e}_y)$$

$$\vec{e}_{\text{右}} = \frac{1}{\sqrt{2}} (\vec{e}_x - i\vec{e}_y)$$

则有

$$\vec{E} = E_x \vec{e}_x + E_y \vec{e}_y = E_{\text{左}} \vec{e}_{\text{左}} + E_{\text{右}} \vec{e}_{\text{右}}$$

$$\Rightarrow E_x = \frac{1}{\sqrt{2}}(E_{\text{左}} + E_{\text{右}}), E_y = \frac{i}{\sqrt{2}}(E_{\text{左}} - E_{\text{右}})$$

瞬时能量密度和能流密度:

$$w = \varepsilon_0 |Re(E)|^2 = \frac{1}{\mu_0} |Re(B)|^2$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = c\varepsilon_0 |Re(E)|^2 \hat{n} = \frac{c}{\mu_0} |Re(B)|^2 \hat{n}$$

周期平均:

$$\langle |Re(E)|^2 \rangle = \frac{1}{2} \vec{E}_0 \cdot \vec{E}_0^*$$

$$\langle |Re(B)|^2 \rangle = \frac{1}{2} \vec{B}_0 \cdot \vec{B}_0^*$$

5.2 电磁波在介质表面的折射和反射

入射、反射、折射的电场分别记作: $\vec{E}, \vec{E}', \vec{E}''$, 界面记作 x 方向。

频率

$$\omega = \omega' = \omega''$$

波矢

$$k_x = k'_x = k''_x$$

角度

$$i = i'$$

$$n \sin i = n' \sin i'$$

振幅

$$\vec{B} = \frac{1}{v} \hat{n} \times \vec{E}$$

垂直线偏振 (s 偏振):

$$\frac{E'_s}{E_s} = \frac{n \cos i - n' \cos i'}{n \cos i + n' \cos i'} = \frac{\sin(i' - i)}{\sin(i' + i)}$$

$$\frac{E''_s}{E_s} = \frac{2n \cos i}{n \cos i + n' \cos i'}$$

平行线偏振 (p 偏振):

$$\frac{E'_p}{E_p} = \frac{n' \cos i - n \cos i'}{n' \cos i + n \cos i'} = \frac{\tan(i - i')}{\tan(i + i')}$$

$$\frac{E''_p}{E_p} = \frac{2n \cos i}{n' \cos i + n \cos i'}$$

proof: 先考虑 s 偏振

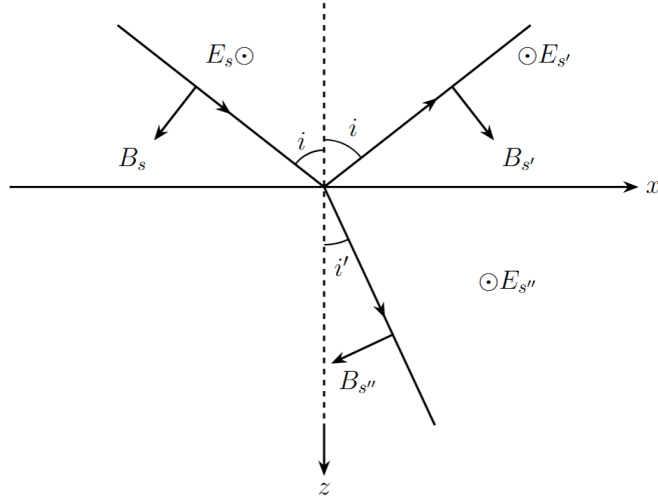


图 1: s 偏振示意图

$\vec{E}$ 法向连续:

$$E_s + E_s' = E_s''$$

$\vec{B}$ 法向连续:

$$B_s \sin i + B_s' \sin i = B_s'' \sin i'$$

$\vec{H}$ 切向连续:

$$\frac{1}{\mu} (B_s \cos i - B_s' \cos i) = \frac{1}{\mu'} B_s'' \cos i'$$

不考虑磁介质:

$$\mu = \mu' = \mu_0, B = \frac{n}{c} E$$

得到

$$nE_s \cos i - nE_s' \cos i = n'E_s'' \cos i'$$

$$E_s + E_s' = E_s''$$

联立解得

$$E_s' = \frac{n \cos i - n' \cos i'}{n \cos i + n' \cos i'} E_s$$

$$E_s'' = \frac{2n \cos i}{n \cos i + n' \cos i'} E_s$$

再考虑 p 偏振:

E 切向连续:

$$E_p \cos i - E_p' \cos i = E_p'' \cos i'$$

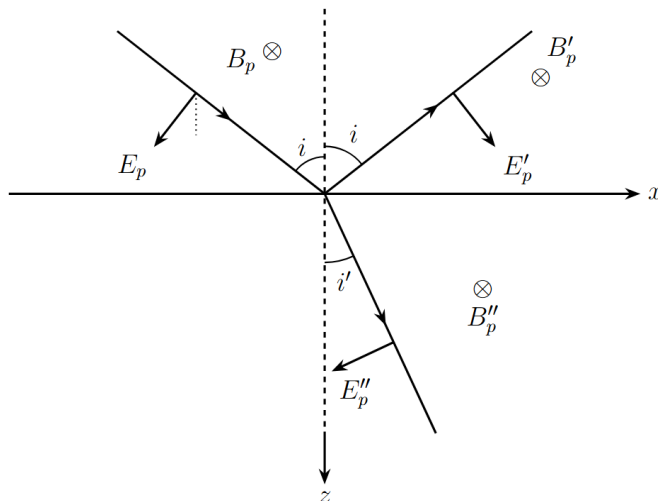


图 2: p 偏振示意图

D 法向连续:

$$\varepsilon_1 (E_p \sin i + E'_p \sin i) = \varepsilon E''_p \sin i'$$

H 切向连续:

$$\frac{1}{\mu_1} (B_p + B'_p) = \frac{B''_p}{\mu_2}$$

联立解得:

$$E'_p = \frac{n' \cos i - n \cos i'}{n' \cos i + n \cos i'} E_p$$

$$E''_p = \frac{2n \cos i}{n' \cos i + n \cos i'} E_p$$

考虑 p 偏振:

$$i = i_B = \arctan \frac{n'}{n}$$

$$\Rightarrow E'_p = 0$$

i_B 称为布儒斯特角

半波损失: 垂直偏振的波从光疏介质入射到光密介质, 反射波存在半波损失。

全反射

$$i = i_c = \arcsin \frac{n'}{n}$$

$$k''_z = i \sqrt{\left(\frac{n}{n'} \sin i\right)^2 - 1} k'' = i\tau$$

5.3 导电介质中的电磁波

$$\vec{j} = \sigma \vec{E}$$

$$\Rightarrow \nabla \times \vec{B} = \mu \sigma \vec{E} + \varepsilon \mu \frac{\partial \vec{E}}{\partial t}$$

认为电荷为零:

$$\begin{aligned} \nabla \cdot \vec{j} &= -\frac{\partial \rho}{\partial t} \\ \Rightarrow \rho &= \rho_0 e^{-\frac{\sigma}{\varepsilon} t} \end{aligned}$$

波动方程:

$$\nabla^2 \vec{E} - \mu \sigma \frac{\partial \vec{E}}{\partial t} - \varepsilon \mu \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

仍然设解:

$$\begin{aligned} \vec{E} &= \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ \Rightarrow -k^2 + i\mu\sigma\omega + \varepsilon\mu\omega^2 &= 0 \end{aligned}$$

1、 $t = 0$ 时刻存在扰动电场，傅里叶展开:

$$\vec{E}_0 = \iiint \vec{\varepsilon}(k) e^{i\vec{k} \cdot \vec{r}} d^3k$$

此时 k 为实数

$$\begin{aligned} \omega &= \frac{-i\mu\sigma \pm \sqrt{-\mu^2\sigma^2 + 4k^2\varepsilon\mu}}{2\varepsilon\mu} = \omega - i\omega_1 \\ \Rightarrow \vec{E} &= \vec{\varepsilon} e^{-\omega_1 t} (k) e^{i(\vec{k} \cdot \vec{r} - \omega_0 t)} \end{aligned}$$

2、真空入射到导体
频率相等:

$$\omega = \omega'$$

此时波矢为复数:

$$k' = k + i\tau$$

实虚部分别对应相等

$$\begin{aligned} k_0^2 - \tau^2 &= \varepsilon\mu\omega^2 \\ k_0\tau &= \frac{1}{2}\mu\sigma\omega \end{aligned}$$

(a) 不良导体

$$\tau \ll k_0$$

$$\Rightarrow k_0 = \sqrt{\varepsilon\mu}\omega, \quad \tau = \frac{1}{2}\sqrt{\frac{\mu}{\varepsilon}}\sigma$$

(b) 良导体

$$\begin{aligned} \tau &\approx k_0 \\ \Rightarrow \tau &= k_0 = \sqrt{\frac{\mu\sigma\omega}{2}} \end{aligned}$$

振幅关系

反射波:

$$\frac{E'_0}{E_0} = \frac{k' - k}{k + k'}$$

透射波:

$$\frac{E''_0}{E_0} = \frac{2k}{k + k'}$$

5.4 波导

之前我们考虑的都是无限延展的平面波，现在来考虑限制在一个空管内的电磁波（波导），在波导内受限的波一般不是横波。假设波导是一个理想导体，即在材料内有 $\vec{E} = 0, \vec{B} = 0$ ，于是波导内的电磁波的边界条件为

$$\begin{aligned} \vec{E}_\tau &= 0 \\ \vec{B}_n &= 0 \end{aligned}$$

考虑沿管轴传播的单色电磁波:

$$\vec{E}(x, y, z, t) = \vec{E}_0(x, y, z, t) e^{i(kz - \omega t)}$$

$$\vec{B}(x, y, z, t) = \vec{B}_0(x, y, z, t) e^{i(kz - \omega t)}$$

在波导内部，电磁场也必须满足 Maxwell 方程组:

$$\begin{aligned} \nabla \cdot \vec{E} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{B} &= \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

根据电磁波的形式，可以看出

$$\begin{aligned} \frac{\partial}{\partial t} &= -i\omega \\ \frac{\partial}{\partial z} &= ik \end{aligned}$$

于是由 Maxwell 方程组，得到

$$\begin{aligned}
\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= i\omega B_z \\
\frac{\partial E_z}{\partial y} - ikE_y &= i\omega B_x \\
ikE_x - \frac{\partial E_z}{\partial x} &= i\omega B_y \\
\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= -\frac{i\omega}{c^2} E_z \\
\frac{\partial B_z}{\partial y} - ikB_y &= -\frac{i\omega}{c^2} E_x \\
ikB_x - \frac{\partial B_z}{\partial x} &= -\frac{i\omega}{c^2} E_y
\end{aligned}$$

可以用只含 E_z, B_z 导数的表达式表示 E_x, E_y, B_x, B_y ：

$$\begin{aligned}
E_x &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial x} + \omega \frac{\partial B_z}{\partial y} \right) \\
E_y &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial y} - \omega \frac{\partial B_z}{\partial x} \right) \\
B_x &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial x} - \frac{\omega}{c^2} \frac{\partial E_z}{\partial y} \right) \\
B_y &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial y} + \frac{\omega}{c^2} \frac{\partial E_z}{\partial x} \right)
\end{aligned}$$

得到关于 E_z, B_z 的方程：

$$\begin{aligned}
\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] E_z &= 0 \\
\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] B_z &= 0
\end{aligned}$$

如果 $E_z = 0$ ，称为 TE 波；如果 $B_z = 0$ ，称为 TM 波；如果 $E_z = B_z = 0$ ，称为 TEM 波。在中空波导中 TEM 波不能发生。

proof：如果 $E_z = 0$ ，由高斯定理

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = 0$$

$B_z = 0$ ，由电磁感应定律

$$\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} = 0$$

散度和旋度都为零，故它可写成一个标量势的梯度，满足 Laplace 方程。但 $\vec{E}$ 的边界条件要求表面是等势面，而 Laplace 方程没有局域极值，因此势是一个常数，故电场为零。

现在来考虑一个具体的例子：考虑矩形波导中的 TE 波，矩形波导的高为 a ，宽为 b 。分离变量

$$B_z(x, y) = X(x)Y(y)$$

代入其满足的偏微分方程：

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} + \left[\left(\frac{\omega}{c} \right)^2 - k^2 \right] XY = 0$$

得到

$$\begin{aligned} \frac{1}{X} \frac{d^2 X}{dx^2} &= -k_x^2 \\ \frac{1}{Y} \frac{d^2 Y}{dy^2} &= -k_y^2 \end{aligned}$$

其中

$$-k_x^2 - k_y^2 + \left(\frac{\omega}{c} \right)^2 - k^2 = 0$$

方程的解为

$$X(x) = A \sin(k_x x) + B \cos(k_x x)$$

边界条件：

$$B_x = 0 \Rightarrow \frac{dX}{dx} = 0, x = 0, a$$

解得

$$A = 0$$

$$k_x = \frac{m\pi}{a}, m = 0, 1, 2 \dots$$

同理

$$k_y = \frac{n\pi}{b}, n = 0, 1, 2 \dots$$

得到

$$B_z = B_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$$

这个解称为 TE_{mn} 模式。

$$k = \sqrt{(\omega/c)^2 - \pi^2 [(m/a)^2 + (n/b)^2]}$$

定义截止频率：

$$\omega_{mn} = c\pi \sqrt{(m/a)^2 + (n/b)^2}$$

要求

$$\omega > \omega_{mn}$$

可以将波矢写作

$$k = \frac{1}{c} \sqrt{\omega^2 - \omega_{mn}^2}$$

相速度

$$v_p = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}}$$

群速度

$$v_g = \frac{d\omega}{dk} = c \sqrt{1 - (\omega_{mn}/\omega)^2}$$

群速度是小于光速的，因为波携带的能量是以群速度传播的。

对于 TE_{mn} 波，有

$$\begin{aligned} B_z &= B_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \\ E_x &= -\frac{i\omega}{(\omega/c)^2 - k^2} \frac{n\pi}{b} B_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \\ E_y &= \frac{i\omega}{(\omega/c)^2 - k^2} \frac{m\pi}{a} B_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \\ B_x &= -\frac{ik}{(\omega/c)^2 - k^2} \frac{m\pi}{a} B_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) \\ B_y &= -\frac{ik}{(\omega/c)^2 - k^2} \frac{n\pi}{b} B_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \end{aligned}$$

共轴传输线有可能传播 TEM 模式波。在一个半径为 a 的长导体线和一个内径为 b 的柱形导体壳所组成的长直共轴传输线中，允许 $E_z = 0, B_z = 0$ 的模式存在，该情形下：

$$k = \frac{\omega}{c}$$

$$cB_y = E_x$$

$$cB_x = -E_y$$

Maxwell 方程组：

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = 0$$

$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = 0$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0$$

$$\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = 0$$

这就是二维真空中的静电场和静磁场方程，解为

$$\vec{E}(r, \phi, z, t) = \frac{A \cos(kz - \omega t)}{r} \vec{e}_r$$

$$\vec{B}(r, \phi, z, t) = \frac{A \cos(kz - \omega t)}{cr} \vec{e}_\phi$$

6 电磁波的激发

现在考虑存在电荷分布与电流分布的情况

6.1 推迟势

Maxwell 方程组：

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{j}$$

电磁势：

$$\vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

得到方程：

$$\nabla^2 \phi + \frac{\partial}{\partial t} (\nabla \cdot \vec{A}) = -\frac{\rho}{\varepsilon_0}$$

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) = -\mu_0 \vec{j}$$

洛伦兹规范：

$$\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0$$

得到方程：

$$\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}$$

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{j}$$

下面来求解 ϕ :

考虑点源的贡献:

$$\begin{aligned}\rho(\vec{r}', t) &= Q(\vec{r}', t) \delta(\vec{r} - \vec{r}') \\ \Rightarrow \nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} &= -\frac{Q(t) \delta(\vec{r} - \vec{r}')}{\varepsilon_0}\end{aligned}$$

定义:

$$\vec{R} = \vec{r} - \vec{r}'$$

$$\phi = \phi(R, t)$$

$$\Rightarrow \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial \phi}{\partial R} \right) - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = -\frac{Q(t) \delta(\vec{R})}{\varepsilon_0}$$

考虑 $R > 0$ 的区域, 令

$$\begin{aligned}\phi(R, t) &= \frac{X(R, t)}{R} \\ \Rightarrow \frac{\partial^2 X}{\partial R^2} - \frac{1}{c^2} \frac{\partial^2 X}{\partial t^2} &= 0 \\ \Rightarrow X(R, t) &= f\left(t - \frac{R}{c}\right) + g\left(t + \frac{R}{c}\right)\end{aligned}$$

取物理解:

$$\phi(R, t) = \frac{f\left(t - \frac{R}{c}\right)}{R}$$

$R = 0$ 处积分:

$$\begin{aligned}\iiint \left(\nabla^2 \frac{f(t)}{R} - \frac{1}{c^2 R} \frac{\partial^2 f}{\partial t^2} \right) d^3 r &= -\frac{Q(t)}{\varepsilon_0} \\ \nabla^2 \frac{f(t)}{R} &= f \nabla^2 \frac{1}{R} + 2 \nabla f \cdot \nabla \frac{1}{R} + \frac{1}{R} \nabla^2 f\end{aligned}$$

只有第一项的积分有贡献:

$$\begin{aligned}f(t) &= \frac{Q(t)}{4\pi\varepsilon_0} \\ \Rightarrow \phi &= \frac{1}{4\pi\varepsilon_0} \frac{Q\left(t - \frac{R}{c}\right)}{R}\end{aligned}$$

叠加原理:

$$\begin{aligned}\phi(\vec{r}, t) &= \frac{1}{4\pi\varepsilon_0} \iiint \frac{\rho\left(\vec{r}', t - \frac{|\vec{r} - \vec{r}'|}{c}\right)}{|\vec{r} - \vec{r}'|} d^3 r' \\ \vec{A}(\vec{r}, t) &= \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}\left(\vec{r}', t - \frac{|\vec{r} - \vec{r}'|}{c}\right)}{|\vec{r} - \vec{r}'|} d^3 r'\end{aligned}$$

可以由势求出场，得到 Jefimenko's 方程：

$$\vec{E}(\vec{r}, t) = \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho(\vec{r}', t')}{R^2} \vec{e}_R + \frac{\dot{\rho}(\vec{r}', t')}{cR} \vec{e}_R - \frac{\ddot{\vec{j}}(\vec{r}', t')}{c^2 R} \right] d^3 r'$$

$$\vec{B}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\ddot{\vec{j}}(\vec{r}', t')}{R^2} + \frac{\dot{\vec{j}}(\vec{r}', t')}{cR} \right] \times \vec{e}_R d^3 r'$$

6.2 谐振电流的电磁场

考虑

$$\vec{j} = \vec{j}_0(\vec{r}) e^{-i\omega t}$$

$$\Rightarrow \nabla \cdot \vec{j} = -i\omega \vec{j}$$

$$\Rightarrow \vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \iiint \frac{\vec{j}_0(\vec{r}') e^{-i\omega \left(t - \frac{R}{c}\right)}}{|\vec{r} - \vec{r}'|} d^3 r'$$

考虑远场 $r \gg r'$ ：

$$\frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{r} + \frac{\vec{r}' \cdot \vec{e}_r}{r^2}$$

$$ik|\vec{r} - \vec{r}'| = ik(r - \vec{e}_r \cdot \vec{r}') = i(kr - \vec{k} \cdot \vec{r}')$$

$$\Rightarrow \vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \iiint \vec{j}_0(\vec{r}') e^{-i\vec{k} \cdot \vec{r}'} d^3 r' \frac{e^{i(kr - \omega t)}}{r} = \vec{A}_0 \frac{e^{i(kr - \omega t)}}{r}$$

$$\vec{A}_0 = \frac{\mu_0}{4\pi} \iiint \vec{j}_0(\vec{r}') e^{-i\vec{k} \cdot \vec{r}'} d^3 r'$$

磁感应强度：

$$\vec{B} = \nabla \times \vec{A} = i\vec{k} \times \vec{A} = (i\vec{k} \times \vec{A}_0) \frac{e^{i(kr - \omega t)}}{r}$$

电场强度：

$$\vec{E} = c\vec{B} \times \vec{e}_r$$

能流密度：

$$\vec{S} = \frac{c}{\mu_0} |\text{Re}(\vec{B})|^2 \vec{e}_r$$

$$\langle \vec{S} \rangle = \frac{c}{2\mu_0} \vec{B} \cdot \vec{B}^* = \frac{c}{2\mu_0 r^2} \vec{B}_0 \cdot \vec{B}_0^*$$

辐射功率：

$$P = \iint \langle \vec{S} \rangle r^2 d\Omega$$

$$\frac{dP}{d\Omega} = \langle \vec{S} \rangle r^2 = \frac{c}{2\mu_0} \vec{B}_0 \cdot \vec{B}_0^*$$

6.3 电偶极辐射

考虑 e 指数因子的展开:

$$e^{-i\vec{k}\cdot\vec{r}'} = 1 - i\vec{k}\cdot\vec{r}'$$

第一项对应电偶极辐射, 矢势为:

$$\vec{A}_0 = \frac{\mu_0}{4\pi} \iiint \vec{j}_0(\vec{r}') d^3r'$$

利用公式:

$$\nabla' \cdot (\vec{j}_0 \vec{r}') = (\nabla' \cdot \vec{j}_0) \vec{r}' + \vec{j}_0$$

对体积积分, 左边为零:

$$\iiint \vec{j}_0 d^3r' = - \iiint (\nabla' \cdot \vec{j}_0) \vec{r}' d^3r'$$

而已知:

$$\begin{aligned} \nabla' \cdot \vec{j}_0 &= i\omega\rho_0 \\ \Rightarrow \iiint \vec{j}_0 d^3r' &= - \iiint i\omega\rho_0 \vec{r}' d^3r' = -i\omega\vec{p}_0 \\ \Rightarrow \vec{A}_0 &= -\frac{i\mu_0}{4\pi}\omega\vec{p}_0 \\ \vec{B} = \nabla \times \vec{A} &= \frac{\mu_0\omega}{4\pi} (\vec{k} \times \vec{p}_0) \frac{e^{i(kr-\omega t)}}{r} \end{aligned}$$

辐射角分布:

$$\begin{aligned} \frac{dP}{d\Omega} &= \frac{c}{2\mu_0} \vec{B}_0 \cdot \vec{B}_0^* = \frac{\mu_0\omega^4}{32\pi^2c} p_0^2 \sin^2\theta \\ P &= \frac{\mu_0\omega^4}{12\pi} p_0^2 \end{aligned}$$

电偶极辐射还可以构建一个电偶极模型来推导得出。想象两个小金属球相距为 d , 并用一根导线相连。在 t 时刻, 上面的小球带电 $q(t)$, 下面的小球带电 $-q(t)$ 。假设驱动电荷通过导线从一端到另一端以角频率 ω 来回运动:

$$q(t) = q_0 \cos(\omega t)$$

产生振荡的电偶极子:

$$\begin{aligned} \vec{p}(t) &= p_0 \cos(\omega t) \hat{z} \\ p_0 &= q_0 d \end{aligned}$$

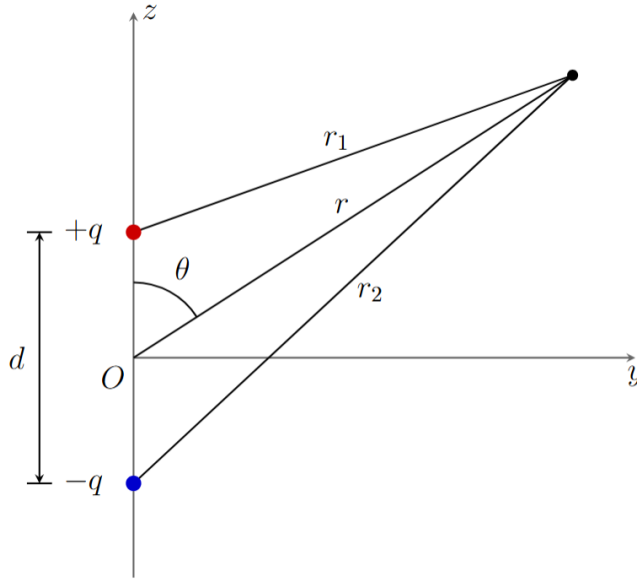


图 3: 振荡电偶极子模型示意图

推迟势公式:

$$\phi(r, t) = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(\vec{r}', t - \frac{|\vec{r} - \vec{r}'|}{c})}{|\vec{r} - \vec{r}'|} d^3r'$$

$$\Rightarrow \phi(\vec{r}, t) = \frac{q_0}{4\pi\epsilon_0} \left\{ \frac{\cos[\omega(t - r_1/c)]}{r_1} - \frac{\cos[\omega(t - r_2/c)]}{r_2} \right\}$$

由余弦定理:

$$r_1 = \sqrt{r^2 - rd \cos \theta + \frac{d^2}{4}}$$

$$r_2 = \sqrt{r^2 + rd \cos \theta + \frac{d^2}{4}}$$

考虑远场近似 $d \ll r$:

$$r_1 \approx r(1 - \frac{d}{2r} \cos \theta)$$

$$r_2 \approx r(1 + \frac{d}{2r} \cos \theta)$$

$$\Rightarrow \cos[\omega(t - \frac{r_1}{c})] \approx \cos[\omega(t - \frac{r}{c}) + \frac{\omega d}{2c} \cos \theta]$$

$$= \cos[\omega(t - \frac{r}{c})] \cos(\frac{\omega d}{2c} \cos \theta) - \sin[\omega(t - \frac{r}{c})] \sin(\frac{\omega d}{2c} \cos \theta)$$

因为

$$\lambda = \frac{2\pi c}{\omega}$$

$$d \ll \lambda \rightarrow d \ll \frac{c}{\omega}$$

$$\Rightarrow \cos[\omega(t - \frac{r_1}{c})] \approx \cos[\omega(t - \frac{r}{c})] - \frac{\omega d}{2c} \cos \theta \sin[\omega(t - \frac{r}{c})]$$

代入得

$$\phi(r, \theta, t) = \frac{p_0 \cos \theta}{4\pi \varepsilon_0 r} \left\{ -\frac{\omega}{c} \sin[\omega(t - \frac{r}{c})] + \frac{1}{r} \cos[\omega(t - \frac{r}{c})] \right\}$$

$$\omega = 0 \rightarrow \phi = \frac{p_0 \cos \theta}{4\pi \varepsilon_0 r^2}$$

考虑辐射区 $r \gg \lambda$:

$$\phi(r, \theta, t) = -\frac{p_0 \omega}{4\pi \varepsilon_0 c} \frac{\cos \theta}{r} \sin[\omega(t - \frac{r}{c})]$$

磁矢势由流过导线的电流确定:

$$I(t) = \frac{dq}{dt} = -q_0 \omega \sin(\omega t)$$

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int_{-d/2}^{d/2} \frac{-q_0 \omega \sin[\omega(t - r'/c)]}{r'} dz \hat{z}$$

一阶近似下, 积分得:

$$\vec{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin[\omega(t - \frac{r}{c})] \hat{z}$$

计算场:

$$\nabla \phi = \frac{\partial \phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \hat{\theta} \approx \frac{p_0 \omega^2}{4\pi \varepsilon_0 c^2} \frac{\cos \theta}{r} \cos[\omega(t - \frac{r}{c})] \hat{r}$$

$$\frac{\partial \vec{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi r} \cos[\omega(t - \frac{r}{c})] (\cos \theta \hat{r} - \sin \theta \hat{\theta})$$

$$\vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi} \frac{\sin \theta}{r} \cos[\omega(t - \frac{r}{c})] \hat{\theta}$$

$$\vec{B} = \nabla \times \vec{A} = -\frac{\mu_0 p_0 \omega^2}{4\pi c} \frac{\sin \theta}{r} \cos[\omega(t - \frac{r}{c})] \hat{\phi}$$

6.4 磁偶极辐射

考虑第二项, 矢势为

$$\vec{A}_0 = -i \frac{\mu_0}{4\pi} \iiint \vec{k} \cdot \vec{r}' \vec{j}_0(\vec{r}') d^3 r'$$

分解被积函数:

$$\vec{r}' \vec{j}_0 = \frac{1}{2} (\vec{r}' \vec{j}_0 - \vec{j}_0 \vec{r}') + \frac{1}{2} (\vec{r}' \vec{j}_0 + \vec{j}_0 \vec{r}')$$

第一项对应磁偶极辐射, 第二项对应电四极辐射

考虑磁偶极辐射, 数学公式:

$$\vec{k} \times (\vec{r}' \times \vec{j}_0) = -\vec{k} \cdot (\vec{r}' \vec{j}_0 - \vec{j}_0 \vec{r}')$$

$$\Rightarrow \vec{A}_1 = i \frac{\mu_0}{4\pi} \vec{k} \times \frac{1}{2} \iiint \vec{r}' \times \vec{j}_0 d^3 r' = \frac{i\mu_0}{4\pi} \vec{k} \times \vec{m}_0$$

磁偶极定义:

$$\vec{m}_0 = \frac{1}{2} \iiint \vec{r}' \times \vec{j}_0 d^3 r'$$

磁感应强度:

$$\vec{B}_{01} = i \vec{k} \times \vec{A}_1 = -\frac{\mu_0}{4\pi} \vec{k} \times (\vec{k} \times \vec{m}_0)$$

$$|\vec{B}_{01}| = \frac{\mu_0 m_0 k^2}{4\pi} \sin \theta$$

辐射角分布:

$$\frac{dP}{d\Omega} = \frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \sin^2 \theta$$

$$P = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}$$

同理，磁偶极辐射也可以构造一个电流环来求解，假设电流环半径为 a ，通有交变电流

$$I(t) = I_0 \cos(\omega t)$$

$$\vec{m}(t) = \pi a^2 I(t) \hat{z} = m_0 \cos(\omega t) \hat{z}$$

$$m_0 = \pi a^2 I_0$$

环上只有流动的电流，不存在静止的电荷分布，因此标势为零。考虑磁矢势:

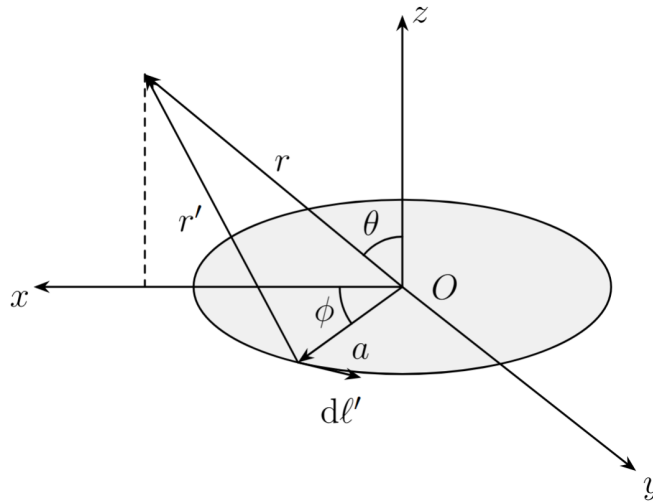


图 4: 振荡磁偶极子模型示意图

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \frac{I_0 \cos[\omega(t - r'/c)]}{r'} d\vec{l}'$$

对于在 x 轴上的 $\vec{r}$, $\vec{A}$ 必定指向 y 方向, 对称分布在 x 轴两边的 x 分量彼此抵消, 于是

$$\begin{aligned}\vec{A}(\vec{r}, t) &= \frac{\mu_0 I_0 a}{4\pi} \hat{y} \int_0^{2\pi} \frac{\cos[\omega(t - r'/c)]}{r'} \cos \phi d\phi \\ \vec{r} &= (r \sin \theta, 0, r \cos \theta) \\ \vec{a} &= (a \cos \phi, a \sin \phi, 0) \\ \Rightarrow r' &= \sqrt{r^2 + a^2 - 2ra \sin \theta \cos \phi}\end{aligned}$$

近似: $a \ll r$, 得到

$$\begin{aligned}r' &\approx r(1 - \frac{a}{r} \sin \theta \cos \phi) \\ \frac{1}{r'} &\approx \frac{1}{r}(1 + \frac{a}{r} \sin \theta \cos \phi) \\ \cos[\omega(t - \frac{r'}{c})] &\approx \cos[\omega(t - \frac{r}{c}) + \frac{\omega a}{c} \sin \theta \cos \phi] \\ &= \cos[\omega(t - \frac{r}{c})] \cos(\frac{\omega a}{c} \sin \theta \cos \phi) - \sin[\omega(t - \frac{r}{c})] \sin(\frac{\omega a}{c} \sin \theta \cos \phi)\end{aligned}$$

近似: $a \ll \lambda$, 得到

$$\cos[\omega(t - \frac{r'}{c})] \approx \cos[\omega(t - \frac{r}{c})] - \frac{\omega a}{c} \sin \theta \cos \phi \sin[\omega(t - \frac{r}{c})]$$

代入积分:

$$\begin{aligned}\vec{A}(r, \theta, t) &= \frac{\mu_0 m_0 \sin \theta}{4\pi r} \left\{ \frac{1}{r} \cos[\omega(t - \frac{r}{c})] - \frac{\omega}{c} \sin[\omega(t - \frac{r}{c})] \right\} \hat{\phi} \\ \omega = 0 &\rightarrow \vec{A}(r, \theta) = \frac{\mu_0 m_0 \sin \theta}{4\pi r^2} \hat{\phi}\end{aligned}$$

辐射区域 $r \gg \lambda$

$$\vec{A}(r, \theta, t) = -\frac{\mu_0 m_0 \omega \sin \theta}{4\pi c r} \sin[\omega(t - \frac{r}{c})] \hat{\phi}$$

计算场:

$$\begin{aligned}\vec{E} &= -\frac{\partial \vec{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2 \sin \theta}{4\pi c r} \cos[\omega(t - \frac{r}{c})] \hat{\phi} \\ \vec{B} &= \nabla \times \vec{A} = -\frac{\mu_0 m_0 \omega^2 \sin \theta}{4\pi c^2 r} \cos[\omega(t - \frac{r}{c})] \hat{\theta}\end{aligned}$$

7 运动电荷的电磁场

7.1 李纳-维谢尔势

考虑点电荷 e 的运动:

$$\vec{r}' = \vec{r}_e(t)$$

电荷分布

$$\rho(\vec{r}', t) = e\delta(\vec{r}' - \vec{r}_e(t))$$

推迟势:

$$\phi = \frac{e}{4\pi\epsilon_0} \iiint \frac{\delta\left(\vec{r}' - \vec{r}_e\left(t - \frac{|\vec{r} - \vec{r}'|}{c}\right)\right)}{|\vec{r} - \vec{r}'|} d^3r'$$

δ 函数积分:

$$\int g(x) \delta(f(x)) dx = \frac{g(x_0)}{f'(x_0)}$$

x_0 满足:

$$f(x_0) = 0$$

引入变换:

$$\vec{r}'' = \vec{r}' - \vec{r}_e\left(t - \frac{|\vec{r} - \vec{r}'|}{c}\right)$$

$$t' = t - \frac{|\vec{r} - \vec{r}'|}{c}$$

$$\vec{R}' = \vec{r} - \vec{r}'$$

我们要把积分元从 d^3r' 换成 d^3r'' , 变换公式:

$$d^3r'' = J d^3r'$$

$$J = \left| \frac{\partial \vec{r}''}{\partial \vec{r}'} \right| = |\nabla' \vec{r}''|$$

下面来求雅可比行列式: (t, r 不变)

$$\nabla' \vec{r}'' = \nabla' \vec{r}' - \nabla' \vec{r}_e(t') = \vec{I} - \nabla' \vec{r}_e(t')$$

$$= \vec{I} - \frac{d\vec{r}_e}{dt'} \nabla' t' = \vec{I} - \vec{v}'(t') \nabla' t'$$

$$\nabla' t' = -\frac{1}{c} \nabla' R' = \frac{\vec{R}'}{cR'}$$

$$\Rightarrow \nabla' \vec{r}'' = \vec{I} - \frac{\vec{v}' \vec{R}'}{R'c}$$

$$|\nabla' \vec{r}''| = 1 - \frac{\vec{v}' \cdot \vec{R}'}{R'c}$$

因此电势为：

$$\phi = \frac{e}{4\pi\epsilon_0} \iiint \frac{\delta(\vec{r}'') d^3r''}{R' \left(1 - \frac{\vec{R}' \cdot \vec{v}'}{R'c}\right)} = \frac{e}{4\pi\epsilon_0} \frac{1}{R^* \left(1 - \frac{\vec{R}^* \cdot \vec{v}^*}{R^*c}\right)}$$

同理，磁矢势为：

$$\vec{A} = \frac{\mu_0}{4\pi} \frac{e\vec{v}^*}{R^* \left(\frac{\vec{R}^* \cdot \vec{v}^*}{R^*c}\right)} = \frac{1}{c^2} \phi \vec{v}^*$$

t^* 是推迟时间，满足：

$$\vec{r}'' = 0$$

$$c(t - t^*) = |\vec{r} - \vec{r}_e(t^*)|$$

7.2 运动电荷产生的电磁场

求出来电势和磁矢势之后就可以求电磁场：

$$\vec{E} = -\nabla\phi - \frac{\partial\vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

（必须抄一句课本原话：原则上把李纳-维谢尔势代入就可以求出来 $\vec{E}$ 和 $\vec{B}$ ，但是这推导远不是直截了当的）

先求电场，规定：

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{e}{s^*}$$

$$\vec{A} = \frac{1}{c^2} \phi \vec{v}^*$$

$$s^* = R^* - \frac{\vec{R}^* \cdot \vec{v}^*}{c}$$

$$R^* = \vec{r} - \vec{r}_e(t^*)$$

考虑函数：

$$f = f(\vec{r}, t^*(\vec{r}, t))$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial t^*} \frac{\partial t^*}{\partial t}$$

$$\nabla f = \nabla_{\star} f + \frac{\partial f}{\partial t^{\star}} \nabla t^{\star}$$

下面计算几个中间结果：

1、 $\frac{\partial R^{\star}}{\partial t^{\star}}$

$$\begin{aligned} \frac{\partial R^{\star}}{\partial t^{\star}} &= \frac{\partial}{\partial t^{\star}} \sqrt{\vec{R}^{\star} \cdot \vec{R}^{\star}} = \frac{1}{2R^{\star}} \frac{\partial}{\partial t^{\star}} (\vec{R}^{\star} \cdot \vec{R}^{\star}) \\ &= \frac{1}{2R^{\star}} 2\vec{R}^{\star} \cdot \frac{\partial \vec{R}^{\star}}{\partial t^{\star}} = -\frac{\vec{R}^{\star} \cdot \vec{v}^{\star}}{R^{\star}} \end{aligned}$$

因为 $\vec{r}$ 固定，所以

$$\frac{\partial \vec{R}^{\star}}{\partial t^{\star}} = -\frac{d\vec{r}_e}{dt^{\star}} = -\vec{v}^{\star}$$

2、 $\frac{\partial t^{\star}}{\partial t}$

已知：

$$t^{\star} = t - \frac{1}{c} |\vec{r} - \vec{r}_e(t^{\star})|$$

对 t 求导：

$$\begin{aligned} \frac{\partial t^{\star}}{\partial t} &= 1 - \frac{1}{c} \frac{\partial R^{\star}}{\partial t^{\star}} \frac{\partial t^{\star}}{\partial t} = 1 + \frac{\vec{R}^{\star} \cdot \vec{v}^{\star}}{R^{\star} c} \frac{\partial t^{\star}}{\partial t} \\ \Rightarrow \frac{\partial t^{\star}}{\partial t} &= \frac{1}{1 - \frac{\vec{R}^{\star} \cdot \vec{v}^{\star}}{R^{\star} c}} = \frac{R^{\star}}{s^{\star}} \end{aligned}$$

3、 $\frac{\partial R^{\star}}{\partial t}$

$$\frac{\partial R^{\star}}{\partial t} = \frac{\partial R^{\star}}{\partial t^{\star}} \frac{\partial t^{\star}}{\partial t} = -\frac{\vec{R}^{\star} \cdot \vec{v}^{\star}}{s^{\star}}$$

4、 $\nabla t^{\star}, \nabla R^{\star}$

求 $\nabla t^{\star}$ 时保持 t 不变：

$$\nabla t^{\star} = -\frac{1}{c} \nabla R^{\star}$$

$$\nabla R^{\star} = \nabla_{\star} R^{\star} + \frac{\partial R^{\star}}{\partial t^{\star}} \nabla t^{\star}$$

代入前面 1 中求过的偏导数，可以解出来

$$\nabla R^{\star} = \frac{\vec{R}^{\star}}{s^{\star}}$$

$$\nabla t^{\star} = -\frac{1}{c} \frac{\vec{R}^{\star}}{s^{\star}}$$

5、 $\nabla_{\star} s^{\star}$

$$s^{\star} = R^{\star} - \frac{1}{c} \vec{R}^{\star} \cdot \vec{v}^{\star}$$

$$\Rightarrow \nabla_{\star} s^{\star} = \frac{\vec{R}^{\star}}{R^{\star}} - \frac{\vec{v}^{\star}}{c}$$

6、 $\frac{\partial s^{\star}}{\partial t^{\star}}$

$$\frac{\partial s^{\star}}{\partial t^{\star}} = -\frac{\vec{R}^{\star} \cdot \vec{v}^{\star}}{R^{\star}} + \frac{\vec{v}^{\star} \cdot \vec{v}^{\star}}{c} - \frac{\vec{R}^{\star} \cdot \vec{a}^{\star}}{c}$$

好了，现在可以求电场了！

$$\begin{aligned} \vec{E} &= -\nabla\phi - \frac{\partial \vec{A}}{\partial t} = -\frac{e}{4\pi\epsilon_0} \left[\nabla \frac{1}{s^{\star}} + \frac{1}{c^2} \frac{\partial}{\partial t} \left(\frac{\vec{v}^{\star}}{s^{\star}} \right) \right] \\ &= -\frac{e}{4\pi\epsilon_0} \left[-\frac{\nabla s^{\star}}{s^{\star 2}} + \frac{1}{c^2} \frac{1}{s^{\star 2}} \left(\frac{\partial \vec{v}^{\star}}{\partial t} s^{\star} - \vec{v}^{\star} \frac{\partial s^{\star}}{\partial t} \right) \right] \\ &= \frac{e}{4\pi\epsilon_0 s^{\star 2}} \left(\nabla s^{\star} + \frac{1}{c^2} \frac{\partial s^{\star}}{\partial t} \vec{v}^{\star} - \frac{1}{c^2} R^{\star} \vec{a}^{\star} \right) \end{aligned}$$

再代入我们已经求过的结果：

$$\vec{E} = \frac{e}{4\pi\epsilon_0 s^{\star 3}} \left\{ \left(1 - \frac{v^{\star 2}}{c^2} \right) \left(\vec{R}^{\star} - \frac{R^{\star}}{c} \vec{v}^{\star} \right) + \frac{1}{c^2} \vec{R}^{\star} \times \left[\left(\vec{R}^{\star} - \frac{R^{\star}}{c} \vec{v}^{\star} \right) \times \vec{a}^{\star} \right] \right\}$$

磁感应强度：

$$\vec{B} = \frac{1}{c} \frac{\vec{R}^{\star}}{R^{\star}} \times \vec{E}$$

7.3 匀速运动电荷的电磁场

对于匀速运动： $\vec{a}^{\star} = 0$

$$\Rightarrow \vec{E} = \frac{e}{4\pi\epsilon_0 s^{\star 3}} \left(1 - \frac{v^2}{c^2} \right) \left(\vec{R}^{\star} - \frac{R^{\star} \vec{v}}{c} \right)$$

考虑场点 $\vec{r}$ 向速度 $\vec{v}$ 方向的投影：

$$\begin{aligned} r_{//} &= \frac{\vec{r} \cdot \vec{v}}{v} \\ r_{\perp} &= |\vec{r} - \vec{r}_{//}| \\ \vec{R}^{\star} - \frac{R^{\star} \vec{v}}{c} &= \vec{r} \\ \Rightarrow R^{\star 2} &= r_{\perp}^2 + \left(\frac{R^{\star} v}{c} + r_{//} \right)^2 \\ s^{\star} &= \frac{1}{\gamma} \sqrt{r_{\perp}^2 + \gamma^2 r_{//}^2} \end{aligned}$$

电场为：

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\gamma e \vec{r}}{\left(r_{\perp}^2 + \gamma^2 r_{//}^2 \right)^{3/2}}$$

磁感应强度为:

$$\vec{B} = \frac{1}{c^2} \vec{v} \times \vec{E}$$

把 $\vec{v}$ 的方向作为极轴:

$$E = \frac{e}{4\pi\epsilon_0 r^2} \frac{\gamma}{(\gamma^2 \cos^2 \theta + \sin^2 \theta)^{3/2}}$$

因为是平方反比关系, 因此匀速运动电荷产生的电磁场不产生辐射。

7.4 运动电荷的辐射功率

考虑有加速度的粒子, 贡献辐射的电场强度为:

$$\vec{E} = \frac{e}{4\pi\epsilon_0 s^{*3}} \frac{\vec{R}^*}{c^2} \times \left[\left(\vec{R}^* - \frac{R^*}{c} \vec{v}^* \right) \times \vec{a}^* \right]$$

$$\vec{B} = \frac{1}{c} \frac{\vec{R}^*}{R^*} \times \vec{E}$$

产生的辐射为

$$\frac{dP}{d\Omega} = w R^{*2} \left(c - \vec{v}^* \cdot \frac{\vec{R}^*}{R^*} \right)$$

$$P = \iint w R^* \left(c - \vec{v}^* \cdot \frac{\vec{R}^*}{R^*} \right) d\Omega$$

能量密度:

$$w = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right) = \epsilon_0 E^2$$

能流密度:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = wc \frac{\vec{R}^*}{R^*}$$

考虑非相对论近似 $v \ll c, \vec{v}/\vec{a}$:

$$\vec{E} = \frac{e}{4\pi\epsilon_0 c^2} \frac{\vec{R}^* \times (\vec{R}^* \times \vec{a}^*)}{R^{*3}}$$

方向角记作 θ

$$E = \frac{ea^* \sin \theta}{4\pi\epsilon_0 c^2 R^*}$$

能量密度:

$$w = \epsilon_0 E^2 = \frac{e^2 a^{*2} \sin^2 \theta}{16\pi^2 \epsilon_0 c^4 R^{*2}}$$

辐射角分布:

$$\frac{dP}{d\Omega} = wc R^{*2} = \frac{e^2 a^{*2} \sin^2 \theta}{16\pi^2 \epsilon_0 c^3}$$

辐射总功率：

$$P = \frac{e^2 a^{\star 2}}{6\pi\epsilon_0 c^3}$$

考虑相对论情形：

1、 $\vec{v} // \vec{a}$

$$E = \frac{1}{4\pi\epsilon_0} \frac{ea^{\star} \sin \theta}{c^2 R^{\star} \left(1 - \frac{v^{\star 2}}{c^2}\right)^3}$$

辐射角分布：

$$\frac{dP}{d\Omega} = \frac{e^2}{16\pi^2\epsilon_0 c^3} \frac{a^{\star 2} \sin^2 \theta}{\left(1 - \frac{v^{\star 2}}{c^2}\right)^5}$$

辐射总功率：

$$P = \frac{e^2}{6\pi\epsilon_0 c^2} \frac{a^{\star 2}}{\left(1 - \frac{v^{\star 2}}{c^2}\right)^3} = \gamma^6 P_{n.r.}$$

考虑极值情况：

$$\varepsilon = \frac{1}{\gamma^2}$$

求导数：

$$\frac{d}{d\theta} \left(\frac{dP}{d\Omega} \right) = 0 \Rightarrow \cos \theta_m = \frac{\sqrt{16 - 15\varepsilon} - 1}{3(1 - \varepsilon/2)} = 1 - \frac{\varepsilon}{8}$$

$$\theta_m = \sqrt{\frac{\varepsilon}{4}} = \frac{1}{2\gamma}$$

2、 $\vec{v} \perp \vec{a}$

$$E^2 = \frac{e^2 \left[\left(1 - \frac{\vec{v}^{\star} \cdot \vec{R}^{\star}}{cR^{\star}}\right) a^{\star 2} - \left(1 - \frac{v^{\star 2}}{c^2} \left(\vec{a}^{\star} \cdot \frac{\vec{R}^{\star}}{R^{\star}}\right)\right) \right]}{16\pi^2\epsilon_0^2 c^4 R^{\star 2} \left(1 - \frac{\vec{v}^{\star} \cdot \vec{R}^{\star}}{cR^{\star}}\right)^6}$$

取 $\vec{a}$ 为 x 轴， $\vec{v}$ 为 z 轴。

$$\vec{R}^{\star} = R^{\star} (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

辐射角分布：

$$\frac{dP}{d\Omega} = \frac{e^2}{16\pi^2\epsilon_0 c^3} \frac{a^{\star 2}}{\left(1 - \frac{v^{\star}}{c} \cos \theta\right)^3} \left[1 - \frac{\sin^2 \theta \cos^2 \phi \left(1 - \frac{v^{\star 2}}{c^2}\right)}{\left(1 - \frac{v^{\star}}{c} \cos \theta\right)^2} \right]$$

辐射总功率：

$$P = \frac{e^2}{6\pi\epsilon_0 c^2} \frac{a^2}{\left(1 - \frac{v^2}{c^2}\right)^2} = \gamma^4 P_{n.r.}$$

一般情形，辐射总功率：

$$P = \frac{e^2}{6\pi\epsilon_0 c^3} (\gamma^6 a_{//}^2 + \gamma^4 a_{\perp}^2) = P_{//} + P_{\perp}$$

或者

$$P = \frac{\mu_0 q^2 \gamma^6}{6\pi c} \left(a^2 - \left| \frac{\vec{v} \times \vec{a}}{c} \right|^2 \right)$$

考虑受力

$$\vec{F} = \frac{d}{dt} (\gamma m_0 \vec{v}) = m_0 \gamma \vec{a} + m_0 \gamma^3 \left(\frac{\vec{a} \cdot \vec{v}}{c} \right) \frac{\vec{v}}{c}$$

$$F_{//} = m_0 \gamma^3 a_{//}$$

$$F_{\perp} = m_0 \gamma a_{\perp}$$

功率比为：

$$\frac{P_{//}}{P_{\perp}} = \left(\frac{\gamma a_{//}}{a_{\perp}} \right)^2 = \left(\frac{F_{//}}{\gamma F_{\perp}} \right)^2$$

7.5 辐射阻尼力

考虑辐射阻尼力做的功：

$$\vec{F}_{\tau} \cdot \vec{v} dt = -dU = -m\tau a^2 dt$$

$$\tau = \frac{e^2}{6\pi\epsilon_0 c^3 m}$$

周期积分：

$$\int_T m\tau \vec{a} \cdot d\vec{v} = \int_T m\tau d(\vec{a} \cdot \vec{v}) - \int_T m\tau \dot{\vec{a}} \cdot \vec{v} dt$$

右边第一项为零

$$\Rightarrow \vec{F}_{\tau} = m\tau \dot{\vec{a}}$$

考虑辐射阻尼力对简谐振子的影响：

$$m\ddot{x} = -m\omega_0^2 x + m\tau \ddot{x} = -m\omega_0^2 x - m\omega_0^2 \tau \dot{x}$$

$$\Rightarrow x(t) = x_0 e^{-\frac{\gamma t}{2}} e^{-i\omega_0 t}$$

$$\gamma = \omega_0^2 \tau$$

傅里叶展开，考虑每个频率的影响：

$$X(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x(t) e^{i\omega t} dt = -\frac{i}{\sqrt{2\pi}} \frac{A_0}{\omega - \omega_0 + \frac{i\gamma}{2}}$$

$$\frac{dP}{d\omega} \propto |X(\omega)|^2 \propto \frac{1}{(\omega - \omega_0)^2 + \frac{\gamma^2}{4}}$$

辐射强度降低为原来的一半时：

$$\omega = \omega_0 \pm \frac{\gamma}{2}$$

8 介质对电磁波的影响

8.1 自由电子对电磁波的散射

动力学方程：

$$m\ddot{\vec{r}} = e\vec{E}_0 e^{-i\omega t}$$

通解为

$$\vec{r} = -\frac{e\vec{E}_0}{m\omega^2} e^{-i\omega t} + \vec{v}_0 t + \vec{r}_0$$

第一项相当于一个偶极子：

$$\vec{p} = \vec{p}_0 E^{-i\omega t} = -\frac{e^2 \vec{E}_0}{m\omega^2} e^{-i\omega t}$$

辐射功率为

$$\frac{dP}{d\Omega} = \frac{e^4 E_0^2}{32\pi^2 \varepsilon_0 c^3 m^2} \sin^2 \alpha$$

考虑入射波 z 方向，散射方向 $\hat{n}$ 在 $x-z$ 平面上，电场方向 $\hat{E}$ 在 $x-y$ 平面上。

$$\hat{n} \cdot \hat{z} = \cos \theta$$

$$\hat{E} \cdot \hat{x} = \cos \phi$$

$$\Rightarrow E \cos \alpha = \vec{E} \cdot \hat{n} = E \sin \theta \cos \phi$$

$$\Rightarrow \frac{dP}{d\Omega} = \frac{e^4 E_0^2}{32\pi^2 \varepsilon_0 c^3 m^2} (1 - \sin^2 \theta \cos^2 \phi)$$

对 ϕ 取平均：

$$\langle \frac{dP}{d\Omega} \rangle = \frac{e^4 E_0^2}{64\pi^2 \varepsilon_0 c^3 m^2} (1 + \cos^2 \theta)$$

总辐射功率：

$$P = \frac{e^4 E_0^2}{12\pi \varepsilon_0 c^3 m^2}$$

可以写作：

$$P = \sigma S_\lambda$$

$$S_\lambda = \frac{1}{2} \varepsilon_0 c E_0^2$$

得到汤姆孙截面：

$$\sigma_T = \frac{e^4}{6\pi\varepsilon_0^2 c^4 m^2} = 6.65 \times 10^{-25} \text{cm}^2$$

8.2 束缚电子对电磁波的散射

此时要考虑回复力与辐射阻尼力，动力学方程：

$$m\ddot{\vec{r}} + m\omega_0^2 \vec{r} - m\tau \dddot{\vec{r}} = e\vec{E}_0 e^{-i\omega t}$$

强迫振动解

$$\vec{r} = \frac{eE_0}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}} e^{-i(\omega t + \delta)}$$

$$\tan \delta = \frac{\omega \gamma}{\omega^2 - \omega_0^2}$$

等效偶极矩：

$$\vec{p} = \frac{e\vec{E}_0}{m} \frac{e^{i\delta}}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}} e^{-i\omega t}$$

仍然将辐射功率表示为：

$$P = \sigma S_\lambda$$

$$\sigma = \sigma_T \frac{\omega^4}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}$$

低频情形：瑞利散射公式

$$\sigma = \sigma_T \left(\frac{\omega}{\omega_0} \right)^4$$

9 相对论

洛伦兹变换：

$$\begin{cases} x' = \gamma(x - vt) \\ y' = y \\ z' = z \\ t' = \gamma(t - \frac{vx}{c^2}) \end{cases}$$

速度合成公式：

$$\begin{cases} u'_x = \frac{u_x - v}{1 - vu_x/c^2} \\ u'_y = \frac{u_y}{\gamma(1 - vu_x/c^2)} \\ u'_z = \frac{u_z}{\gamma(1 - vu_x/c^2)} \end{cases}$$

9.1 4-矢量

洛伦兹变换矩阵：

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

4-矢量满足：

$$a'^{\mu} = \sum_{\nu=0}^3 \Lambda_{\nu}^{\mu} a^{\nu}$$

变换：

$$\begin{cases} a'^0 = \gamma(a^0 - \beta a^1) \\ a'^1 = \gamma(a^1 - \beta a^0) \\ a'^2 = a^2 \\ a'^3 = a^3 \end{cases}$$

逆变矢量：

$$a^{\mu} = (a_0, a_1, a_2, a_3)$$

协变矢量：

$$a_{\mu} = (-a_0, a_1, a_2, a_3)$$

变换关系为：

$$a_{\mu} = \sum_{\nu=0}^3 g_{\mu\nu} a^{\nu}$$

其中 $g_{\mu\nu}$ 是闵可夫斯基度规：

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

标量积:

$$a_\mu b^\mu = a^\mu b_\mu = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3$$

$$\begin{cases} a^\mu a_\mu > 0 & \text{类空} \\ a^\mu a_\mu = 0 & \text{类光} \\ a^\mu a_\mu < 0 & \text{类时} \end{cases}$$

9.2 速度

固有时:

$$d\tau = \frac{dt}{\gamma}$$

平常速度:

$$\vec{u} = \frac{d\vec{r}}{dt}$$

固有速度:

$$\vec{\eta} = \frac{d\vec{r}}{d\tau} = \gamma \vec{u}$$

4-速度:

$$\boxed{\eta^\mu = \frac{dx^\mu}{d\tau} = \gamma(c, \vec{u})}$$

$$\eta^0 = \frac{dx^0}{d\tau} = \gamma c$$

$$\vec{\eta} = \frac{d\vec{r}}{d\tau} = \gamma \vec{u}$$

变换关系:

$$\begin{pmatrix} \eta'^0 \\ \eta'^1 \\ \eta'^2 \\ \eta'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \eta^0 \\ \eta^1 \\ \eta^2 \\ \eta^3 \end{pmatrix}$$

$$\eta^\mu = \gamma(c, \vec{u})$$

$$\eta_\mu = \gamma(-c, \vec{u})$$

$$\eta^\mu \eta_\mu = \gamma^2(-c^2 + u^2) = -c^2$$

9.3 能动量

$$\vec{p} = m\vec{\eta} = \gamma m \vec{u}$$

$$E = \gamma m c^2$$

能量动量 4-矢量:

$$p^\mu = m\eta^\mu = \gamma m(c, \vec{u}) = \left(\frac{E}{c}, \vec{p} \right)$$

$$p_\mu = m\eta_\mu = \gamma m(-c, \vec{u}) = \left(-\frac{E}{c}, \vec{p} \right)$$

标量积:

$$p^\mu p_\mu = -m^2 c^2$$

$$E^2 - p^2 c^2 = m^2 c^4$$

9.4 力

平常力:

$$\vec{F} = \frac{d\vec{p}}{dt} = \gamma m \vec{a} + \gamma^3 m \frac{\vec{u} \cdot \vec{a}}{c^2} \vec{u}$$

力的变换:

$$\begin{cases} F'_x = \frac{F_x - \beta(\vec{u} \cdot \vec{F})/c}{1 - \beta u_x/c} \\ F'_y = \frac{F_y}{\gamma(1 - \beta u_x/c)} \\ F'_z = \frac{F_z}{\gamma(1 - \beta u_x/c)} \end{cases}$$

闵可夫斯基力:

$$K^\mu = \frac{dp^\mu}{d\tau}$$

$$\vec{K} = \gamma \vec{F}$$

$$K^0 = \frac{1}{c} \frac{dE}{d\tau} = \frac{\gamma}{c} \frac{dE}{dt}$$

洛伦兹力是一种平常力:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

标量积:

$$K^\mu K_\mu = \frac{1 - \beta^2 \cos^2 \theta}{1 - \beta^2} F^2$$

9.5 加速度

$$\vec{a} = \frac{d\vec{u}}{dt}$$

固有加速度:

$$\boxed{\alpha^\mu = \frac{d\eta^\mu}{d\tau} = \frac{d^2 x^\mu}{d\tau^2}}$$

$$\alpha^0 = c \frac{d\gamma}{d\tau} = \gamma^4 \frac{\vec{u} \cdot \vec{a}}{c}$$

$$\vec{\alpha} = \gamma^2 \vec{a} + \gamma^4 \frac{\vec{u} \cdot \vec{a}}{c^2} \vec{u}$$

$$\alpha^\mu = \left(\gamma^4 \frac{\vec{u} \cdot \vec{a}}{c}, \gamma^2 \vec{a} + \gamma^4 \frac{\vec{u} \cdot \vec{a}}{c^2} \vec{u} \right)$$

标量积:

$$\alpha^\mu \alpha_\mu = \gamma^6 \left(a^2 - \frac{1}{c^2} |\vec{u} \times \vec{a}|^2 \right)$$

$$\eta^\mu \alpha_\mu = 0$$

牛顿第二定律:

$$K^\mu = m \alpha^\mu$$

9.6 电磁场

电磁场的相对论变换:

$$\left\{ \begin{array}{l} E'_x = E_x \\ E'_y = \gamma(E_y - vB_z) \\ E'_z = \gamma(E_z + vB_y) \\ B'_x = B_x \\ B'_y = \gamma(B_y + vE_z/c^2) \\ B'_z = \gamma(B_z - vE_y/c^2) \end{array} \right.$$

$$\vec{E} \cdot \vec{B} = \vec{E}' \cdot \vec{B}'$$

$$E^2 - c^2 B^2 = E'^2 - c^2 B'^2$$

电磁场张量：

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

对偶张量：

$$G^{\mu\nu} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{pmatrix}$$

$$F^{\mu\nu} F_{\mu\nu} = 2 \left(B^2 - \frac{E^2}{c^2} \right)$$

$$G^{\mu\nu} G_{\mu\nu} = 2 \left(-B^2 + \frac{E^2}{c^2} \right)$$

$$F^{\mu\nu} G_{\mu\nu} = -\frac{4}{c} \vec{E} \cdot \vec{B}$$

均为洛伦兹不变量。

9.7 电流密度

$$\vec{J} = \rho \vec{u} = \gamma \rho_0 \vec{u}$$

电流密度 4-矢量：

$$J^\mu = \rho_0 \eta^\mu = (c\rho, \vec{J})$$

连续性方程：

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

4-矢量形式：

$$\frac{\partial J^\mu}{\partial x^\mu} = 0$$

Maxwell 方程组：

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \mu_0 J^\mu$$

$$\frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0$$

或者可以写作：

$$\frac{\partial F_{\mu\nu}}{\partial x^\lambda} + \frac{\partial F_{\nu\lambda}}{\partial x^\mu} + \frac{\partial F_{\lambda\mu}}{\partial x^\nu} = 0$$

闵可夫斯基力：

$$K^\mu = qF^{\mu\nu}\eta_\nu$$

9.8 相对论势

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

4-矢量势：

$$A^\mu = \left(\frac{\phi}{c}, \vec{A} \right)$$

电磁场张量与 4-矢量势的关系：

$$F^{\mu\nu} = \frac{\partial A^\nu}{\partial x_\mu} - \frac{\partial A^\mu}{\partial x_\nu}$$

洛伦茨规范：

$$\nabla \cdot \vec{A} = -\frac{1}{c^2} \frac{\partial \phi}{\partial t}$$

$$\frac{\partial A^\mu}{\partial x^\mu} = 0$$

9.9 规范不变性

$$\psi_e \rightarrow \psi'_e = e^{-iq\Lambda/\hbar}$$

$$\vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla\Lambda$$

$$\phi \rightarrow \phi' = \phi - \frac{\partial \Lambda}{\partial t}$$

电磁场中的薛定谔方程：

$$i\hbar \frac{\partial \psi_e}{\partial t} = \left[\frac{1}{2m} \left(-i\hbar \nabla - q\vec{A} \right)^2 + q\phi \right] \psi_e$$

狄拉克方程：

$$\boxed{(\gamma^\mu p_\mu - mc) \psi_e = 0}$$

ψ_e 是一个四分量波函数，称作**旋量**。 γ^μ 是四个 4×4 狄拉克矩阵，可以用 Pauli 矩阵表示：

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\gamma^0 = i \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

满足反对易关系：

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I$$